

SparseDiff: Baseline vs Novel Algorithm  
Ego Dataset — 100 Epochs

| Metric             | Baseline<br>(Original) | Novel<br>(Ours) | Change  |
|--------------------|------------------------|-----------------|---------|
| Degree (MMD) ↓     | 0.0084                 | 0.0126          | -49.9%  |
| Clustering (MMD) ↓ | 0.0648                 | 0.1146          | -76.9%  |
| Orbit (MMD) ↓      | 0.0291                 | 0.0361          | -24.0%  |
| Spectral (MMD) ↓   | 0.0164                 | 0.0124          | +24.0%  |
| FID ↓              | 221.54                 | 257.10          | -16.0%  |
| RBF MMD ↓          | 0.0809                 | 0.1845          | -127.9% |
| Num Nodes W1 ↓     | 9.4901                 | 9.3973          | +1.0%   |
| Edge Types TV ↓    | 0.0070                 | 0.0118          | -68.1%  |
| Disconnected (%) ↓ | 49.01                  | 63.58           | -29.7%  |
| Mean Components ↓  | 2.1722                 | 3.1788          | -46.3%  |
| Max Components     | 13.00                  | 24.00           | +84.6%  |

Green = improvement | Yellow = marginal | Red = regression  
↑ = higher is better | ↓ = lower is better | MMD = Maximum Mean Discrepancy